

MA341

APPLIED ODE'S

Homework 4.1

1. In the following questions you are given a homogeneous ODE together with one of its solutions y_1 . Use Wronskian and Abel's formula to find y_2 .

(a) $xy'' - (x+1)y' + y = 0$, $y_1 = e^x$

Solution: We need to find any linearly independent y_2 , we do this using the Abel's formula

$$W[y_1, y_2](x) = Ce^{-\int -\frac{x+1}{x} dx} = Ce^{(x+\ln x)} = Cxe^x,$$

where for our needs, we can choose any $C \neq 0$. We choose $C = -1$. We next use the other formula of Wronskian and solve

$$W_{1,2}(x) = e^x(y_2' - y_2) = -xe^x.$$

We actually need to solve the first order ODE $y_2' - y_2 = -x$, which we do with the help of integration factor $\mu = e^{-\int dx} = e^{-x}$:

$$\begin{aligned}(e^{-x}y_2)' &= e^{-x}(y_2' - y_2) = -xe^{-x} \\ \Rightarrow y_2 &= -e^x \int xe^{-x} dx = -e^x \left(-xe^{-x} + \int e^{-x} dx \right) = \\ &= -e^x (-xe^{-x} - e^{-x} + C) = (1+x) - Ce^x\end{aligned}$$

We can choose the constant C again, since any linearly independent y_2 will work for us, this time we choose $C = 0$ to get $y_2 = x + 1$.

(b) $x^2y'' - 4xy' + 6y = 0$, $y_1 = x^2$

Solution:

$$W[y_1, y_2](x) = Ce^{-\int -\frac{4}{x} dx} = Ce^{4\ln x} = Cx^4$$

Since any value of C works, let's choose $C = 1$.

$$W[y_1, y_2](x) = \begin{vmatrix} x^2 & y_2 \\ 2x & y_2' \end{vmatrix} = x^2y_2' - 2xy_2 = x^4$$

So we solve

$$y_2' - \frac{2}{x}y_2 = x^2$$

with the help of $\mu = e^{-\int \frac{2}{x} dx} = e^{-2\ln x} = \frac{1}{x^2}$

Thus,

$$\begin{aligned}\left(\frac{1}{x^2}y_2\right)' &= \frac{1}{x^2}\left(y_2' - \frac{2}{x}y_2\right) = 1 \\ \Rightarrow y_2 &= x^2 \int dx = x^2(x+C) = x^3 + Cx^2\end{aligned}$$

We can choose $C = 0$ to get $y_2 = x^3$.

(c) $xy'' + (5x - 1)y' - 5y = 0$, $y_1 = 5x - 1$

Solution:

$$W[y_1, y_2](x) = Ce^{-\int \frac{5x-1}{x} dx} = Ce^{-\int 5 - \frac{1}{x} dx} = Ce^{-(5x - \ln x)} = Cxe^{-5x}$$

Choose $C = -25$.

$$W[y_1, y_2](x) = \begin{vmatrix} 5x-1 & y_2 \\ 5 & y_2' \end{vmatrix} = (5x-1)y_2' - 5y_2 = -25xe^{-5x}$$

So we solve

$$y_2' - \frac{5}{5x-1}y_2 = -\frac{25xe^{-5x}}{5x-1}$$

with the help of

$$\mu = e^{-\int \frac{5}{5x-1} dx} = e^{-\ln(5x-1)} = \frac{1}{5x-1}$$

to get

$$\begin{aligned} \left(\frac{1}{5x-1} y_2 \right)' &= \frac{1}{5x-1} \left(y_2' - \frac{5}{5x-1} y_2 \right) = -\frac{25xe^{-5x}}{(5x-1)^2} \\ \Rightarrow y_2 &= -25(5x-1) \int \frac{xe^{-5x}}{(5x-1)^2} dx \\ &= (5x-1) \underbrace{\int \frac{\overbrace{-5e^{-5x}}^{f'} \overbrace{(5x-1)}^g - \overbrace{5}^{g'} \overbrace{e^{-5x}}^f}{\underbrace{(5x-1)^2}_{g^2}} dx}_{(f/g)'} = (5x-1) \left(\frac{e^{-5x}}{(5x-1)} + C \right) \end{aligned}$$

Choose $C = 9$ to get $y_2 = e^{-5x}$.

(d) $xy'' + (1 - 2x)y' + (x - 1)y = 0$, $y_1 = e^x$

Solution:

$$W[y_1, y_2](x) = Ce^{-\int \frac{1-2x}{x} dx} = Ce^{-(\ln x - 2x)} = C \frac{e^{2x}}{x}$$

Choose $C = 1$

$$W[y_1, y_2](x) = \begin{vmatrix} e^x & y_2 \\ e^x & y_2' \end{vmatrix} = e^x (y_2' - y_2) = \frac{e^{2x}}{x}$$

Thus, we solve $y_2' - y_2 = \frac{e^x}{x}$ with the help $\mu = e^{-x}$, to get

$$\begin{aligned} (e^{-x} y_2)' &= e^{-x} (y_2' - y_2) = \frac{1}{x} \\ \Rightarrow y_2 &= e^x \int \frac{1}{x} dx = e^x (\ln x + C) \end{aligned}$$

Choose $C = 0$ to get $y_2 = e^x \ln x$

2. Find a general solution for the following ODEs:

(a) $6y'' - y' - y = 0$

Solution:

$$y = c_1 e^{x/2} + c_2 e^{-x/3}$$

(b) $y'' - 2y' - 2y = 0$

Solution:

$$y = c_1 e^{(1+\sqrt{3})x} + c_2 e^{(1-\sqrt{3})x}$$

(c) $y'' - 2y' + 6y = 0$

Solution:

$$y = c_1 e^x \cos \sqrt{5}x + c_2 e^x \sin \sqrt{5}x$$

(d) $y'' + 5y' = 0$

Solution:

$$y = c_1 + c_2 e^{-5x}$$

(e) $y'' - 8y' + 16y = 0$

Solution:

$$y = c_1 e^{4x} + c_2 x e^{4x}$$

3. Consider the following initial value problem

$$\begin{cases} y'' - y' - 2y = 0 \\ y(0) = \alpha \\ y'(0) = 2 \end{cases}$$

(a) Find solution to the problem.

Solution: General solution

$$y = c_1 e^{-x} + c_2 e^{2x}$$

using IC we get

$$\begin{cases} c_1 + c_2 = \alpha \\ -c_1 + 2c_2 = 2 \end{cases}$$

adding both equations gives $3c_2 = \alpha + 2$, hence $c_2 = \frac{\alpha+2}{3}$, consequently $c_1 = \alpha - c_2 = \frac{2(\alpha-1)}{3}$, and we got a private solution:

$$y = \frac{2(\alpha-1)}{3} e^{-x} + \frac{\alpha+2}{3} e^{2x}$$

- (b) What values of α gives $\lim_{x \rightarrow \infty} y(x) = 0$.

Solution: If $\alpha = -2$, then

$$y = -2e^{-x} \rightarrow 0 \quad \text{as } x \rightarrow \infty.$$

4. Use D'Alembert Order Reduction Method to find solution to the following ODEs.

- (a) Find y_1 by yourself, then reduce the order and solve the resulting ODE to find y_2 , the general and finally the private solution.

$$\begin{cases} 9y'' + 6y' + y = 0 \\ y(0) = 3 \\ y'(0) = 0 \end{cases}$$

Solution: First we consider characteristic polynomial $9r^2 + 6r + 1 = (1 + 3r)^2$, which has the single root $r = -\frac{1}{3}$, and we got $y_1 = e^{-x/3}$. Assume now solution of the form $y = vy_1 = ve^{-x/3}$, differentiate it and substitute it into the equation:

$$\begin{aligned} y' &= v'e^{-x/3} - \frac{1}{3}ve^{-x/3} = \left(v' - \frac{1}{3}v\right)e^{-x/3} \\ y'' &= v''e^{-x/3} - \frac{2}{3}v'e^{-x/3} + \frac{1}{9}ve^{-x/3} = \left(v'' - \frac{2}{3}v' + \frac{1}{9}v\right)e^{-x/3}, \end{aligned}$$

and

$$\begin{aligned} 0 &= 9\left(v'' - \frac{2}{3}v' + \frac{1}{9}v\right)e^{-x/3} + 6\left(v' - \frac{1}{3}v\right)e^{-x/3} + ve^{-x/3} \\ 0 &= 9\left(v'' - \frac{2}{3}v' + \frac{1}{9}v\right) + 6\left(v' - \frac{1}{3}v\right) + v \\ &= 9v'' - 6v' + v + 6v' - 2v + v \\ &= 9v''. \end{aligned}$$

Thus, we solve $v'' = 0$, which has the general solution $v = ax + b$. Thus, the solution is

$$y = vy_1 = (ax + b)e^{-x/3}.$$

We now need to determine a, b using ICs. From the first IC

$$y(0) = (a \cdot 0 + b)e^{-0/3} = b = 3,$$

we have

$$y = (ax + 3)e^{-x/3}.$$

Now, differentiate

$$y' = -\frac{1}{3}(ax + 3)e^{-x/3} + ae^{-x/3}$$

and use the second IC:

$$y'(0) = -\frac{1}{3}(a \cdot 0 + 3)e^{-0/3} + ae^{-0/3} = a - 1 = 0$$

So we got $a = 1$ and finally

$$y = (x + 3)e^{-x/3}.$$

(b) You are going to solve the following third order ODE

$$\begin{cases} y''' - 3y'' + 3y' - y = 0 \\ y(0) = 1 \\ y'(0) = 1 \\ y''(0) = 1 \end{cases}.$$

First, use a known private solution $y_1 = e^x$ (verify it is indeed a solution!) to reduce its order, then solve resulting 2nd order ODE. Use it to find the solution of the original problem.

Solution: Assume solution of the form $y = vy_1 = ve^x$, then

$$\begin{aligned} y' &= v'e^x + ve^x = (v' + v)e^x \\ y'' &= v''e^x + 2v'e^x + ve^x = (v'' + 2v' + v)e^x \\ y''' &= v'''e^x + 3v''e^x + 3v'e^x + ve^x = (v''' + 3v'' + 3v' + v)e^x, \end{aligned}$$

and

$$\begin{aligned} 0 &= (v''' + 3v'' + 3v' + v)e^x - 3(v'' + 2v' + v)e^x + 3(v' + v)e^x - ve^x \\ 0 &= (v''' + 3v'' + 3v' + v) - 3(v'' + 2v' + v) + 3(v' + v) - v \\ &= v'''. \end{aligned}$$

Thus, we solve $v''' = 0$. Integration gives $v'' = C_1$, another integration gives $v' = C_1x + C_2$. Integrating it once again we have

$$v = C_1 \frac{x^2}{2} + C_2x + C_3,$$

thus,

$$y = ve^x = \left(C_1 \frac{x^2}{2} + C_2x + C_3 \right) e^x.$$

We now use ICs.

$$y(0) = C_3 = 1.$$

Next,

$$y'(0) = \left(C_1 \frac{x^2}{2} + C_2x + 1 \right) e^x + (C_1x + C_2) e^x \Big|_{x=0} = 1 + C_2 = 1,$$

and we got $C_2 = 0$. Finally,

$$y''(0) = \left(C_1 \frac{x^2}{2} + 1 \right) e^x + 2C_1xe^x + C_1e^x \Big|_{x=0} = 1 + C_1 = 1,$$

and we got $C_1 = 0$.

Therefore

$$y = e^x$$